

Brandon Silva

Links: <https://github.com/brandons209> <https://bsilva.info>
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Summary

Experienced AI and software engineer with a wide range of skills; quick to learn, great problem solver, and experienced in multiple programming languages.

Technical Skills

- Proficient in Java, C, C++, SQL
- Expert in Python
- Git and GitHub
- Package testing and deployment
- Linux and Windows OS
- Linux Command Line
- Docker
- Kubernetes
- AWS and Azure
- Database Management (MySQL)
- Microsoft Office
- Machine Learning (ML)
- Deep Learning (DL) using PyTorch and Keras
- Computer Vision
- Transformers - LLMs
- Data Science Tools (pandas, numpy, matplotlib, scipy etc.)
- Big Data Analysis
- MLOPs (Snowflake, Power BI)
- APIs (FastAPI, PySpark, Flask WebServers)

Experience

Florida Solar Energy Center (FSEC)
1679 Clearlake Rd, Cocoa, FL 32922

October 2020 - Present

Research Assistant, Systems Administrator, Database Engineer

- Developed AI models for predicting faults in photovoltaic systems utilizing IV curves with 90% accuracy
- Built software to automate data collection, analysis, monitoring, and reporting from multiple sources
- Designed and managed databases for storage of research data
 - Relational (MySQL) and Time Series databases (OSI-PI)
- Managed multiple Windows, Linux, and web servers for lab use

University of Florida
1889 Museum Rd, Gainesville, FL 32606

August 2021 - July 2024

Graduate Research Assistant

- Developed AI Transformer models for patient outcome prediction in ICUs
 - Realtime prediction of Delirium, Sepsis, Mortality, and Acuity with over ~75% accuracy
- Mentored peers on utilizing HPC resources for research
- Collaborated with other researchers on computer vision projects; pain prediction and mobility detection using infrared and depth cameras
- Created grant and fellowship submissions for funding

Orange County Academy
18415 11th Ave, Orlando, FL 32833

July 2018 – July 2021

IT, Network and Systems Administrator (Volunteering)

- Developed and implemented school-wide networking infrastructure; switches, servers, routers, and computer labs
- Managed Active Directory, Windows and Linux servers, school data, and G-Suite services
- Provided technical support to staff for laptops, computers, and internet
- Setup workstations, printers, and other peripheral devices for staff and students

Center for Research in Computer Vision (CRCV) at UCF

September 2018 - December 2019

4328 Scorpius St. Suite 245, Orlando, FL 32816

Undergraduate Research Assistant

- One of 10 participants selected for the National Science Foundation backed Research for Undergraduates program in 2019
- Conducted computer vision research alongside PhD students
 - Tracking multiple fast flying UAVs from another UAV for targeting
 - Object and action recognition using UAV footage
- Instructed peers on Linux command line tools, PyTorch, Python, and AI fundamentals
- Hosted workshops on interfacing with High Performance Computers

Education

Bachelors of Science in Computer Engineering

University of Central Florida (UCF)
Orlando, FL 32816

August 2021

Minors:

- Mathematics
- Intelligent Robotics Systems

3.97 GPA, Summa Cum Laude

Lead Scholar Designation

Masters of Science in Computer Science

University of Florida (UF)
Gainesville, FL 32601

December 2023

3.66 GPA

Projects and Publications

- [PyPVRPM](#) - Photovoltaic Reliability and Performance Model in Python
 - Simulates PV systems over their lifetime to estimate costs, profits, and viability of large solar panel power plants using NREL's SAM software
- [Multi-dimensional patient acuity estimation with longitudinal EHR](#)
 - Processes large amounts of electronic health records for patients to predict mortality in ICU
- [Dynamic Delirium Prediction in the Intensive Care Unit using Machine Learning on Electronic Health Records](#)
 - Predict onset of Delirium for patients every 12 hours using AI
- [APRICOT-Mamba: Acuity Prediction in Intensive Care Unit \(ICU\)](#)
 - Utilizes MAMBA and Transformer models for predicting acuity in 4 hour windows
- **MAC** - Modular Autonomous Cart that Assists with Intra-home Object Transportation
 - Senior Design Project for achieving Bachelors in Computer Engineering
- **AI Lecturer** - Designed and hosted numerous workshops to teach students introduction to AI and ML
- **Homelab** - Runs a small home server including NAS with ZFS RAID, web servers, home automation and networking
- **Robotics** - Participated in FIRST Robotics in highschool, robotics senior design project in college, and achieved a minor in Intelligent Robotics Systems